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RESEARCH ARTICLE

A Deep Learning Convolutional Neural Network for Antenna Near-Field Prediction and Surrogate Modeling

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ABSTRACT This study investigates the use of deep learning techniques for building a generalized surrogate model that can accurately and very efficiently predict antenna performance parameters. Notably, we focus on applications where a substantial amount of simulation time is required and prior data is available for deep learning use. Specifically, for these applications, we introduce deep learning models that efficiently and reliably model the near-field of the antenna. These models, in turn, accurately predict far-field properties and essential antenna metrics, such as the reflection coefficient. To demonstrate the efficiency of our method, the widely used rectangular patch antenna is considered, encompassing variations in several important geometrical parameters, dielectric constant, and frequency. Based on our results, the proposed model, once trained, is over 200 times faster than conventional full-wave simulations with a nominal average root mean square error (RMSE) of 0.0174 in predicting all the necessary antenna parameters, such as resonant frequency, radiation pattern, and directivity.

INDEX TERMS Antennas, electromagnetics, deep learning, surrogate modeling.

I. INTRODUCTION

The modeling and prediction of electromagnetic properties through classical numerical methods, such as the finite element method (FEM), method of moment (MoM), or finite difference time domain (FDTD) is inarguably time-consuming, memory-intensive, and computationally demanding. This is particularly evident in scenarios where repetitive calculations are required or when dealing with electrically large structures. To address this problem, surrogate modeling, also known as, meta-modeling, or reduced modeling, a technique for approximating the behavior of complex models with a computationally inexpensive model has been developed. Notably, several different surrogate modeling algorithms have been introduced, such as the kriging [1], co-kriging [2], artificial

neural networks [3], gaussian process regression [4], support vector regression [5], radial basis functions [6] that have been applied in a wide range of problems, spanning from antennas [7], [8], [9], [10], circuits [11], microwave imaging [12], to the general fields of electromagnetics [13], geoscience [14] and chemistry [15].

Recently, there has been a significant interest in developing surrogate models for antenna and reflectarray design problems [16], [17], [18], [19], [20], [21], [22], [23], where automated optimization is reinforced by surrogate models towards a faster design. In such processes, the learning can either occur offline [24], or in real-time [25] during the optimization. In fact, parallelization of the training step has further provided an opportunity for a broader use through a distributed calculation across many users and computers. Notably, these developments are conceptually related to cloud-oriented artificial intelligence-enabled technologies,

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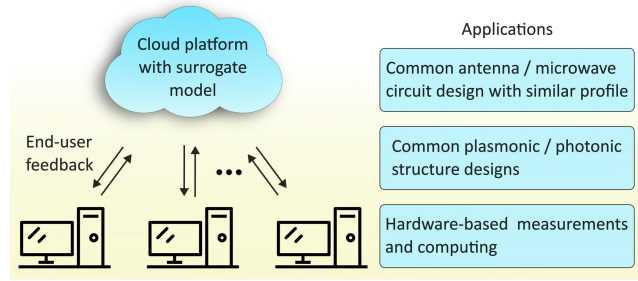


FIGURE 1. Conceptual surrogate modeling framework for data-driven and cloud-supported design.

e.g., Generative Pre-trained Transformers (GPTs) [26]. Such cloud-based surrogate models (see Fig. 1) could be used to build commercial antenna libraries, (e.g., [27]) with fast antenna performance estimation capabilities.

In this work, we investigate a deep learning convolutional neural network for antenna design problems via a novel approach that utilizes the near fields of the antenna. Notably, a severe limitation in prior deep learning-based surrogate models has been the limited number of output parameters that these models can predict (see Table 1), and the lack of physics in the proposed methodologies when training the corresponding models. For example, a network trained using only the reflection coefficient, gain and radiation pattern might not capture important near-field phenomena, such as surface currents, internal resonances, scan blindness due to surface waves, etc. Our method addresses both challenges. Specifically, using near-field data, we can accurately model all the necessary far-field parameters, such as directivity, gain, radiation pattern, and side lobe level for the entire 3-D space using the traditional near-to-far-field transformation, [28]. Also, near-field data can be used to calculate other important antenna characteristics such as coupling [29], cross-polarization generation [30], as well as to conduct antenna diagnostics in general [31]. Overall, utilizing near-field data allows for a more comprehensive analysis, capturing the full characterization of the antenna design. Moreover, this approach also has potential broader applications beyond radiating structures, extending to resonant structures that do not radiate. Notably, deep learning techniques based on near-field data have been demonstrated in electric motor drive design [32], high-frequency plasmonics [8], and nanophotonics [33], as summarized in Table 2.

To properly utilize our near-field data and train our models, well-established image recognition techniques are utilized.¹ To demonstrate the efficiency of our method, a traditional microstrip line fed patch antenna is chosen. Notably, the training time and accuracy of our method are investigated and all the necessary antenna parameters, such as the reflection coefficient, directivity, cross-polarization, and 3-D radiation

¹Image recognition and prediction techniques, such as convolution neural networks (CNNs), are mature technologies that have shown high reliability (e.g., [44]) and integration into commercial products.

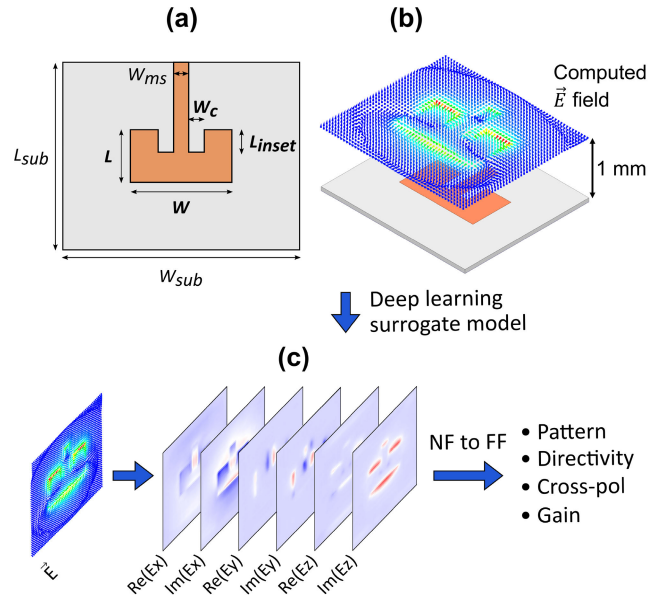


FIGURE 2. (a) An example of the geometry search space (variable parameters highlighted using bold letters) chosen for the antenna modeling process, (b) demonstration of the near E-field vector distribution used to calculate the desired far-field data, (c) pictorial representation of mapping the required near-field components into images used by our proposed CNN-based deep learning approach.

pattern are examined for a wide range of geometrical parameters, dielectric characteristics, and frequencies.

In what follows, Section II presents our method in detail, and Section III demonstrates its efficiency. Finally, Section IV presents our conclusions.

II. CONVOLUTIONAL NEURAL NETWORK BASED NEAR-FIELD MODEL

As briefly discussed in Section I, to properly utilize the near-field data and train our deep learning model, well-established image processing techniques are utilized. Namely, a convolution neural network (CNN) is utilized that handles the complex electric near-field components as images² with the goal to identify the appropriate antenna shape, dielectric constant (ϵ_r), and frequency of operation (see Fig. 2). Notably, unlike traditional image processing applications (e.g., [45], [46], [47], [48]), in our approach, a very fine resolution is required to accurately map every value of the near-field to a grid-point of the discretized image.

CNN-based modeling of the antenna using near-field is broadly related to image processing. In an increasing order of resolution requirement, prior applications are classified as (1) whole image classification [45], (2) labeling of a portion of the image (i.e., box object detection) [46], (3) key-point detection [47], and local correspondence [48]. From the former to the latter, there is an increasing need for resolving features of decreasing sizes.

Here, the goal is to develop a neural network that can map the antenna's shape, dielectric constant, and frequency

²CNNs are specialized neural network architectures capable of handling images with image-features in multiscale sizes.

TABLE 1. Antenna modeling approaches with deep learning and their scope of input and output.

Reference	Application	Input	Output	Deep learning framework
Xiao et al., [34] (2018)	Forward modeling of antennas	3 geometrical variables, 1 dielectric constant point, 1 frequency point	S_{11} , gain, radiation pattern (limited to two principal planes)	Multi-layer perceptron (MLP)
Stankovic et al., [35] (2021)	Forward modeling of antennas	4 geometrical variables, 1 frequency point	Forward gain	MLP
Kapetanakis et al., [36] (2018)	Inverse modeling of a loop antenna	EM field intensity at N points at a specific wavelength	Antenna radius	MLP
Wang et al., [37] (2012)	Inverse modeling of a patch antenna	3 substrate parameters, and frequency of operation	2 geometrical variables	MLP
Kim et al., [38] (2007)	Forward modeling of antennas	7 geometrical variables	Resistance and reactance	MLP
Gupta et al., [39] (2023)	Inverse modeling of antennas	S_{11} vs. frequency at 81 points	Pixelated microstrip antenna	MLP and CNN
Piltan et al., [40] (2024)	Forward modeling of antennas	3 geometrical variables, 7 frequency points, and 13 radiation directions	Gain	MLP, SVRM, GPR, EL, M2LP
This work	Forward modeling of antennas	64×64 spatial points (for 4 geometrical variables), 21 dielectric constants representing homogeneous substrates, 51 frequency points	Near/far-field, and S_{11}	U-net and MLP

TABLE 2. Deep learning-based EM forward modeling approaches utilizing near-field data.

Reference	Domain	Input	Output	Deep learning framework	Training time (per epoch)	Hardware utilized	Prediction time
Khan et al., [41] (2019)	Low-frequency EM	2-D geometry and current	Magnetic near-field for a coil, a transformer, and an interior permanent magnet.	CNN	10 minutes	GPU	20-30 milliseconds
Wiecha et al., [42] (2019)	Nano-photonics	3-D geometry	Electric polarization density inside nanostructures	CNN	2-10 minutes	GPU	3-6 milliseconds
Wu et al., [8] (2021)	Plasmonics	2-D Geometry (3 input parameters)	Near/far-field EM resonances	MLP	-	-	-
Shao et al., [43] (2022)	Microwave breast imaging	2-D geometry	Scattered near-field from 2-D dielectric breast phantoms	CNN	1.2 minutes	GPU	-
This work	Antennas	2-D geometry, dielectric constant, frequency	Near/far-field, and S_{11} for common antennas	U-net and MLP	45-270 minutes	CPU	7 milliseconds

to complex two-dimensional (2-D) E-field components. This relates to a much finer resolution (as compared to the above applications), where each grid point corresponds to a pixel value in the field image, and this operation needs to be done for six field variables using the same network [see Fig. 2(c)]. For this reason, the CNN architecture is a good choice for building our neural network to map such high-dimensional, and multi-scale input and output.

A. NEF-NET FOR ANTENNA NEAR-FIELD MAPPING

To enable the multiscale-sized modeling that we discussed above, we adopt a pioneering strategy inspired by semantic

segmentation techniques [49]. This strategy leverages convolution filters and skip connections to capture the multiscale features of a field map. Originally designed for fine resolution mapping and feature extraction, we adopt this approach in our problem for mapping the geometry of the modeled antenna to its corresponding near-field maps. Namely, to evaluate the near-field data, we consider a hypothetical closed surface surrounding the modeled antenna. This surface can take various forms, such as spherical, parallelepipedal, cubical, and so on. Notably, for microstrip antennas over a ground plane that predominantly radiate only towards the half-spherical plane, this hypothetical surface does not need to be closed. In such

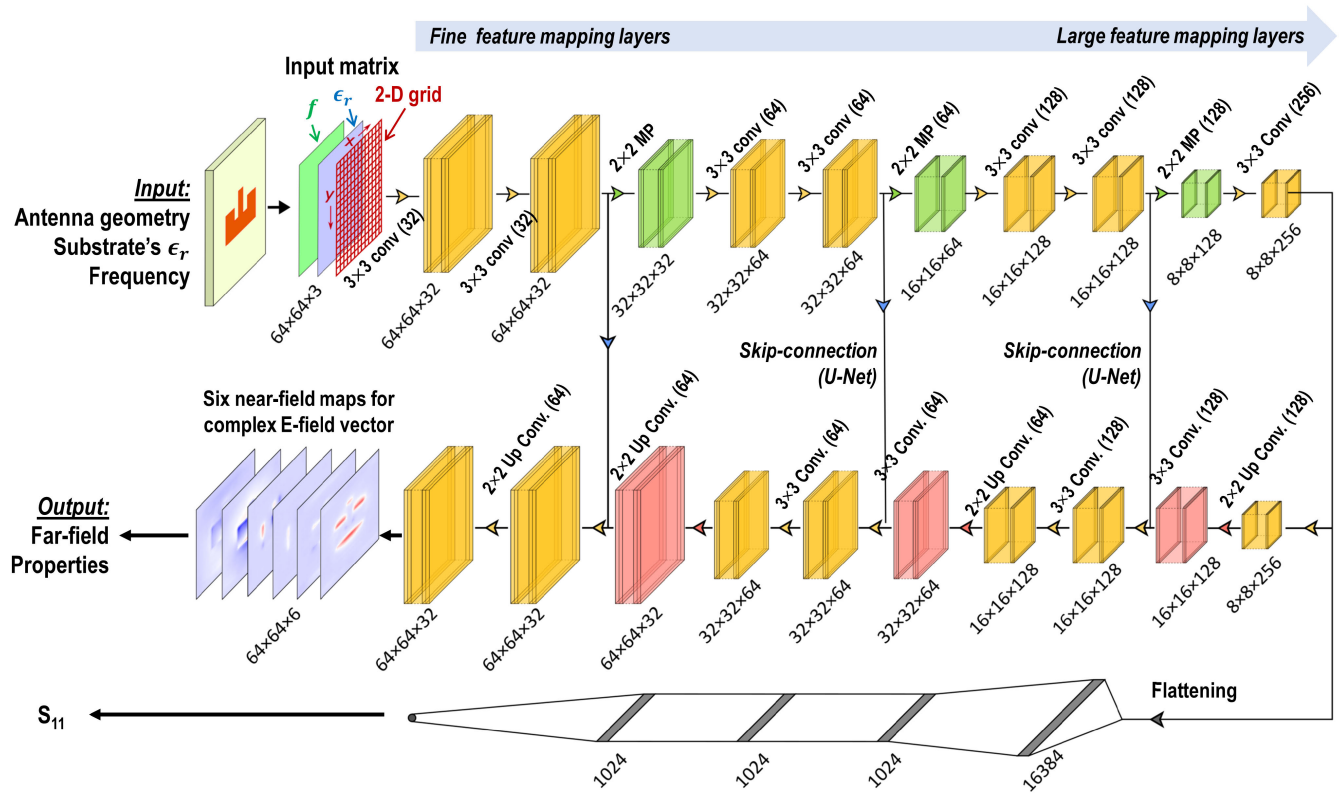


FIGURE 3. The proposed *NeF-Net* architecture for generalized antenna EM modeling.

cases, a single plane is sufficient for accurately evaluating the antenna's performance.³ For example, in the case of the patch antenna shown in Fig. 2(a), we only consider the upper plane shown in Fig. 2(b). Furthermore, we integrate frequency and dielectric substrate information within the network. In this work, we refer to this neural network as the *NeF-Net* (Near-Field Neural Network). This innovative framework combines the power of convolutional neural networks with the ability to accurately capture antenna characteristics and near-field behavior. The architecture of *NeF-Net* is shown in Fig. 3. In what follows, a three-point justification for choosing this architecture is given.

(1) In image processing, to capture features of varying scales, a range of convolution filter sizes and max-pooling layers is needed. Convolution filters enable the extraction of features at different scales. For instance, a 2×2 convolution filter with a single stride calculates a weighted average over four-pixel points, shifting the map one pixel at a time. This allows for the extraction of finer features from the field map. On the other hand, larger filters, such as those with a size of 3×3 , compute a weighted average over nine points, enabling the extraction of larger features while skipping smaller ones. Notably, each convolution stage incorporates multiple convolution filters, with up to 128 filters employed. This abundance of filter weights ensures that there

³The interested reader is referred to reviewing the uniqueness theorem and surface equivalence principle, [28].

TABLE 3. Patch antenna parameters.

Parameter	Lower limit	Upper limit
L	13.72 mm	20.58 mm
W	26.31 mm	39.47 mm
L_{inset}	6.82 mm	9.23 mm
W_c	3.1 mm	5.17 mm
ϵ_r	2	4
Frequency	4 GHz	6 GHz

are sufficient parameters available for optimization, facilitating the modeling of complex features within the antenna system.

(2) Max-pooling layers play a crucial role in reducing the spatial dimension of a feature map by selecting the maximum value within a filter neighborhood. This operation, combined with subsequent convolution filters applied on the reduced feature maps, enables the extraction of larger features within the image. In summary, the initial convolution and max-pool layers focus on extracting finer features, while the subsequent layers with smaller feature maps focus on extracting larger features (refer to Fig. 3). To ensure separate and independent paths for the multiscale layers, skip connections (originally introduced in [49]) are utilized. These skip connections facilitate the merging of these processes into a single map, improving the flow of gradients during training. This, in turn, leads to faster and more effective training of the network.

(3) The *NeF-Net* also incorporates an up-convolutional layer, which expands the feature maps to a higher-dimensional output. This expansion, for example, can transform a feature map from $8 \times 8 \times 256$ to $16 \times 16 \times 128$, as depicted in Fig. 3. The purpose of this up-convolutional layer is to establish a one-to-one mapping between the antenna grid and the field plane. With this architecture, the total number of trainable parameters in the neural network is 20.8 million. This significant parameter count ensures sufficient complexity to accurately map the six output field variables. The implementation of the neural network is carried out in the Python environment using the TensorFlow library [50] and its associated tool environment. Finally, the hyper-parameters of *NeF-Net* were tuned manually, a common practice followed in well-known deep learning models (e.g., Resnet in [51], AlexNet in [52], LeNet-5 in [53]).

B. NEF-NET INPUT AND OUTPUT ARCHITECTURE

Our model considers various geometrical parameters of a rectangular patch antenna, including L (length), W (width), L_{inset} , and W_c , as outlined in Table 3. Additionally, the substrate's dielectric constant (ϵ_r) and a specific frequency range are taken into account. Furthermore, the substrate thickness is assumed to be constant at 1.575 mm, based on commercially available substrates. Notably, the chosen variations and parameter ranges in this study are tailored for developing a surrogate model for C-band antennas, with applications in Wi-Fi, 5G, and satellite communications. However, the selection of the variable space and limits can be customized based on the desired applications, allowing flexibility for different antenna designs.

The antenna metal shape is represented as a binary image, where grid points within the metal region are assigned unit values, while non-metal regions are assigned zero. This binary image grid aligns with the plane over which near-field predictions are sought. Additionally, two additional layers of 2-D images or matrices are used to input frequency and dielectric constant information, respectively (as shown in the input matrix from Fig. 3). By incorporating these parameters in the input matrix, the *NeF-Net* seamlessly integrates geometry, frequency, and dielectric constant, generating an output that is a function of all these factors.

The integration of parameters is achieved through convolution operations applied to the input matrix during the initial stage of the *NeF-Net*. Moreover, by employing a 2-D matrix representation of the dielectric constant, variable dielectric constants can be accommodated in different regions of the substrate in future works, if necessary. This flexible integration allows for comprehensive modeling and analysis of antenna behavior in relation to frequency and dielectric constant variations.

The network output consists of a complex E-field vector distribution, which comprises six numerical values at each grid point. These values represent the real and imaginary parts of the E_x , E_y , and E_z field components, as shown in Fig. 2(c).

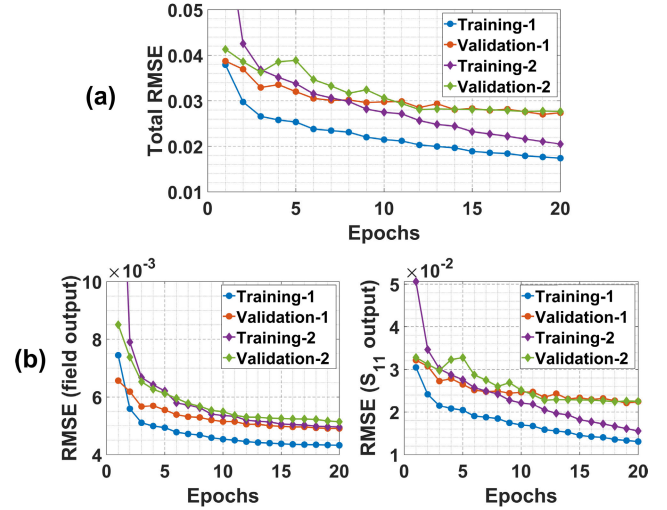


FIGURE 4. (a) Total training and validation RMSE loss for the two training configurations mentioned in Section III-A, (b) RMSE loss components for the two outputs, namely, electric field output, and S_{11} output.

To predict S_{11} (dB), the *NeF-Net* is designed to map the antenna image, dielectric constant, and frequency inputs to a single value. To account for the influence of all input parameters, the first half of the *NeF-Net* architecture (as depicted in Fig. 3) is utilized. To obtain a single output value, parallel flattening layers are applied, creating a parallel feed-forward neural network with a single-point output. Notably, adding more neurons at the output layer can enable the prediction of additional parameters, e.g., real and imaginary parts of the input impedance, enabling a more comprehensive antenna analysis.

III. NEF-NET TRAINING, RESULTS, AND DISCUSSION

In the previous section, we provided an overview of the *NeF-Net* framework, including its inputs and outputs. In this section, we present specific training cases and examine the accuracy of *NeF-Net* for predicting the near-field, far-field, and S_{11} of several microstrip patch antennas.

A. NEF-NET TRAINING

To train the *NeF-Net*, an input dataset containing near E-field data and S_{11} values is utilized. Specifically, our training used 800 randomly selected antenna geometries (using Latin Hypercube Sampling algorithm), 21 variations of uniformly spaced dielectric constants (homogeneous substrates), and 51 variations of uniformly spaced excitation frequencies selected from the parameter ranges outlined in Table 3. In this process, 856,800 training instances were used for training. To enable the automated generation of this dataset, a MATLAB script was used for generating this dataset using the FEM solver in ANSYS HFSS. The electric field vector associated with the near-field was calculated across a 64×64 grid on a plane with a dimension of $1.5\lambda_g \times 1.5\lambda_g$ ($\lambda_g = 42.43$ mm, considered as the wavelength in the dielectric at 5 GHz frequency for a substrate with $\epsilon_r = 2$) positioned 1 mm above the patch antenna, as shown in Fig. 2(b). The

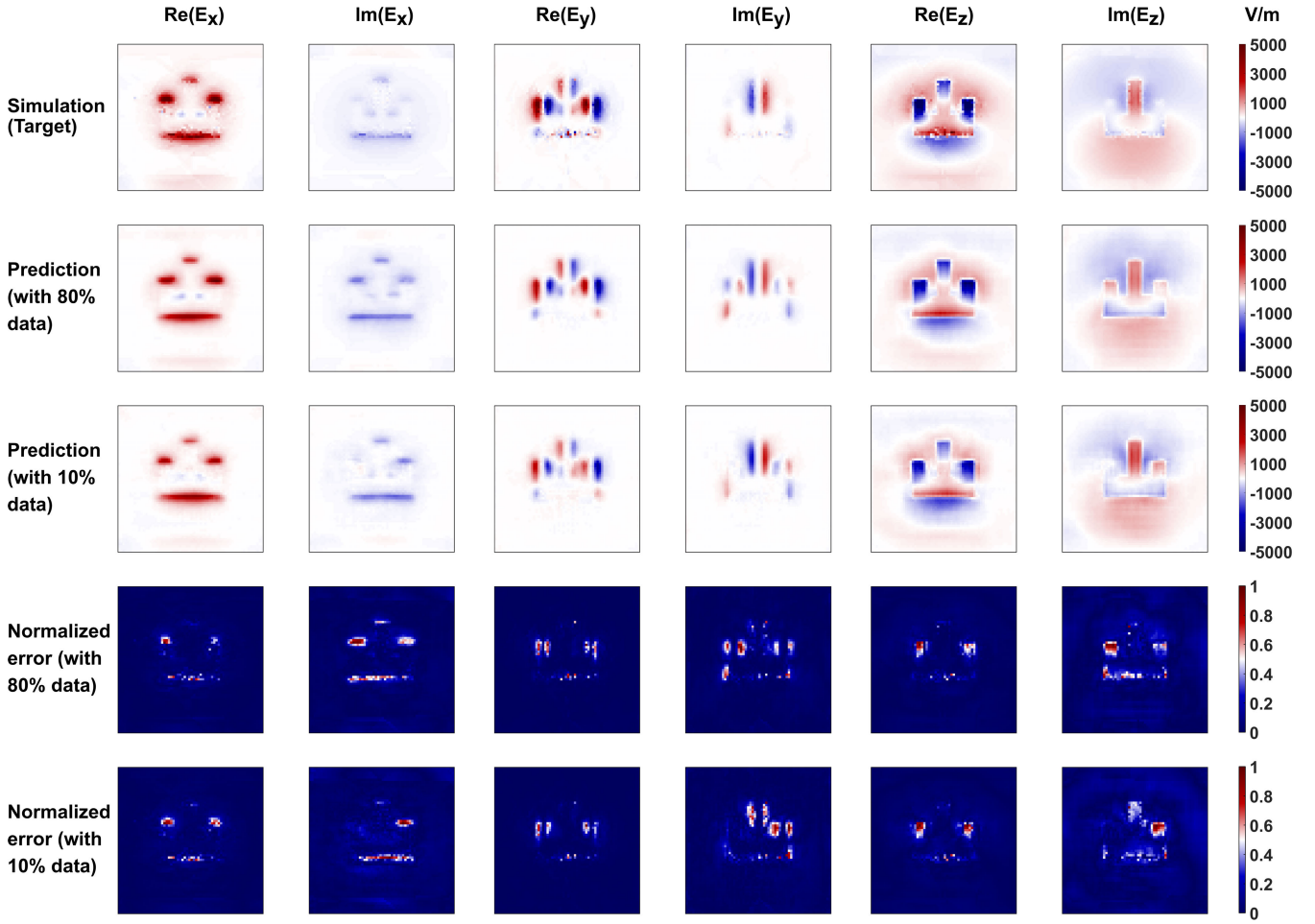


FIGURE 5. Predicted distribution of electric field components for antenna 2 at the resonant frequency under two training configurations are shown, and compared to the field distributions found using full-wave simulation. Normalized estimation error (i.e., residual error) is also shown.

spacing between the field points is sufficiently small (e.g., $\approx \lambda_g/50$ or smaller) to resolve the near-field variations near the edges of the antenna. The selection of this plane allows the consideration of the most important (in terms of intensity) E-fields in the vicinity of the patch antenna so that far-field predictions can be accurately made. A prior study on this method for a rectangular patch antenna is presented in [30].

NeF-Net weights and biases are initialized randomly from a uniform distribution. To train the ANN, we employ the stochastic gradient descent with an adaptive momentum estimation (i.e., Adam [54]) optimization algorithm. Initially, a learning rate of 0.0012 is used, and it is dynamically adjusted during the training using a step decay schedule. Specifically, the learning rate is reduced by a factor of 1/3 after every 3 epochs to prevent oscillations in the loss curve. We use a composite RMSE in the loss calculation, which includes near-field RMSE, and S_{11} RMSE with equal weights. The following definition of RMSE is used,

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (1)$$

TABLE 4. Training summary.

	Dataset generation time (hours)	<i>NeF-Net</i> training time (hours)	RMSE
10% data	13.9	15	0.0205
80% data	111.9	90	0.0174

where, N is the number of samples, and y_i , and $\hat{y}_i$ are the ground truth, e.g., simulated E-field and $S_{11}(\text{dB})$, and the model's estimate, e.g., predicted E-field and $S_{11}(\text{dB})$, respectively.

The generation time for the complete data set is 139.9 hours utilizing four Intel Xeon Gold 6258R CPU-based machines. The time taken to generate the training data is based on a significant reduction applied during the parallel [55] nature of the task. We note that the dataset generated for this study is rigorous so that we can test the effect of data-set size. A comparison is made with the use of 10% data training and 80% data training in the results presented in this paper. Table 4 summarizes the data generation and training times for this work.

TABLE 5. *NeF-Net* test antennas.

	L (mm)	W (mm)	L_{inset} (mm)	W_c (mm)	ϵ_r
Antenna 1	14.8	28.5	8.1	3.1	3.66
Antenna 2	14.96	25.3	6.51	6.51	2.2
Antenna 3	16.5	28	8.37	4.32	2.8

Initially, we train the network for 20 epochs using 80% of the training instances and test (or validate) using the remaining instances. In this configuration, training time was approximately 4.5 hours per epoch and 90 hours for 20 epochs to reach a total RMSE error, $RMSE_{\text{total}}$, of 0.0174. The evolution of RMSE loss with the increasing number of epochs is shown in Fig. 4. Notably, training was performed in batches of 50 samples to limit memory usage, and to avoid over-fitting [56]. To understand the effect of smaller datasets for training and validation, a second instance of dataset selection and training was pursued using 10% of the total data for training and 1% data for validation. Notably, the validation data is different from the training data in each training. With a reduced dataset the training time was 45 minutes for one epoch and 15 hours for 20 epochs, which provided an $RMSE_{\text{total}}$ of 0.0205.

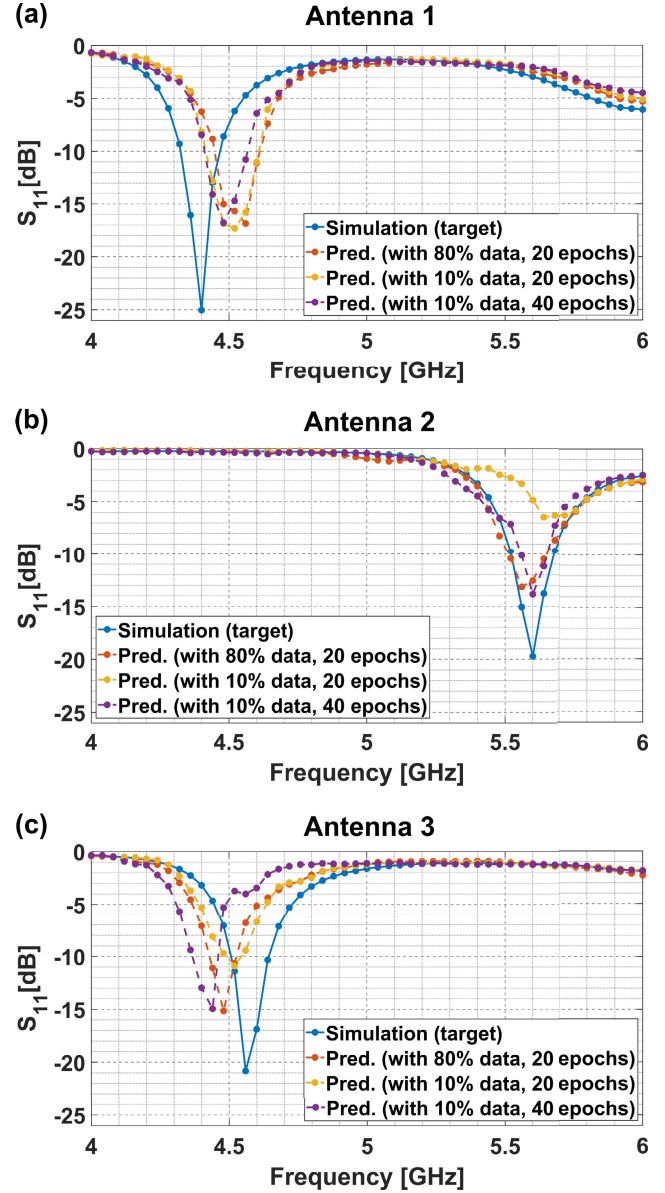
For ease of handling, the field data is represented using a 3-D tensor or 2-D frames containing six channels, where channels represented the 2-D data for each of the following variables: $Re(E_x)$, $Im(E_x)$, $Re(E_y)$, $Im(E_y)$, $Re(E_z)$, $Im(E_z)$. The other output, i.e., S_{11} (dB) is formatted using a 1-D array. Finally, before calculating the RMSE, both input and output tensors are scaled between 0 and 1 (i.e., normalized RGB image format), following the recommendations in [57] to reduce numerical instability in training.

B. NEAR-FIELD PREDICTION

To compare the near-field prediction of the *NeF-Net* with full-wave simulations, the antenna 2 in Table 5 is chosen. Specifically, the predicted x , y , and z components of the electric field are generated using the *NeF-Net* at an operating frequency of 5.65 GHz, and compared with the full-wave simulation in Fig. 5. The normalized error is also shown in this figure. Notably, *NeF-Net* took approximately 7 milliseconds to compute the antenna near-zone electric field and S_{11} at a single frequency. In the case where the *NeF-Net* is trained using 80% of the available data, the near-field RMSE error is 0.0234. On the other hand, when the *NeF-Net* is trained using 10% of the available data, the near-field RMSE error is 0.0379. Therefore, the predictions obtained from our *NeF-Net* demonstrate significant accuracy, even when using small dataset configurations. Section III-D discusses the effect of the near-field errors on our far-field calculations.

C. S_{11} PREDICTION

To test the capability of *NeF-Net* in predicting S_{11} , we perform a 51-point (uniformly spaced) frequency sweep in the frequency range of 4 to 6 GHz for all three antennas tabulated

**FIGURE 6.** S_{11} prediction performance of the proposed ANN (for both training configurations mentioned in Section III-A) over a frequency band for the three test designs tabulated in Table 5.

in Table 5. *NeF-Net*'s prediction of S_{11} vs. frequency curves are shown in Fig. 6 for the three designs, with an RMSE error of 0.17, 0.06, and 0.16, respectively, when the *NeF-Net* was trained using 80% of the available data and 20 epochs, 10% of the available data and 20 epochs, and 10% of the available data and 40 epochs. Table 6 summarizes the estimated resonant frequencies and corresponding errors for the three antennas when compared to full-wave simulation. Notably, an error in predicting the resonant frequency in the order of 1.43% is observed when the *NeF-Net* is trained using 10% of the available data and 40 epochs [see Fig. 6(b)]. Nevertheless, the error remains within the range of uncertainty of dielectric constant in typical antenna measurements of this type and the prediction is still useful for practical applications. The S_{11} -sweep and the electric field prediction time was

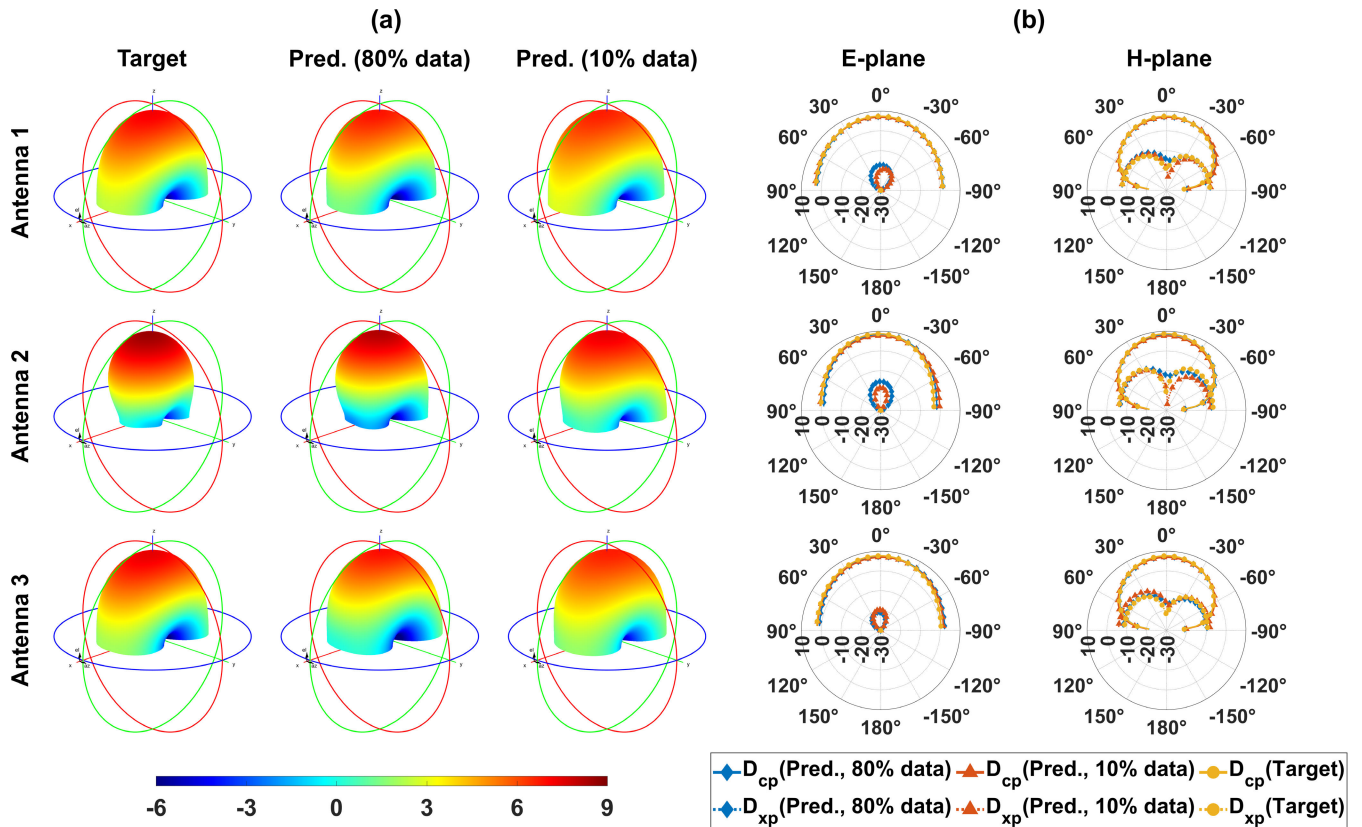


FIGURE 7. (a) 3-D polar plot of total directivity (in dB), and (b) far-field radiation patterns, e.g., directivity in dB along E-plane and H-plane, both co and cross-polarization components (marked with solid lines, and dashed lines, respectively), computed using the near-field predicted by the proposed ANN, and using the near-field data generated by ANSYS HFSS (e.g., target), are shown here for test antenna 1 (at 4.52 GHz), test antenna 2 (at 5.64 GHz), and test antenna 3 (at 4.52 GHz), as mentioned in Table 5.

TABLE 6. Resonant frequency for the three test designs of Table 5.

(a) Resonant frequency ^a (GHz)					
Full-wave	<i>NeF-Net</i>				
	Prediction (80% data)	Prediction (10% data) ^b	Error (80% data)	Error (10% data)	
Antenna 1	4.4	4.52	4.52	2.72%	2.72%
Antenna 2	5.6	5.56	5.68	0.71%	1.43%
Antenna 3	4.56	4.48	4.52	1.75%	0.87%

^a Resonant frequency is estimated by selecting the median frequency point among three lowest S_{11} (dB) frequency points.

^b 20 epoch results are reported.

between 0.36–0.37 seconds (i.e., ≈ 7 milliseconds for each frequency point) which translates to an acceleration by two orders of magnitude when compared to the full-wave solution (Table 9).

D. FAR-FIELD ESTIMATION USING THE PREDICTED NEAR-FIELD DISTRIBUTION

We next estimate the far-field radiation pattern of the antennas based on the predicted near-field. Near-field models are quite useful for this prediction since the entire 3-D patterns can be predicted with a high resolution without the need to conduct

training on extensive far-field datasets. Notably, the near-to-far-field transformation is conducted by calculating the equivalent magnetic surface current distribution $\vec{M}(x,y)$ using the equivalence theorem [28].

Fig. 7(a) shows the total 3-D directivity patterns, and Fig. 7(b) shows the directivity of co- and cross-pol components for both the E- and H-plane of the patch antenna. As it is shown in this figure, our *NeF-Net* based far-field 3-D patterns and the far-field patterns evaluated based on the near-fields directly obtained by full-wave antenna simulations are in very good agreement. Table 7 summarizes the comparison of maximum directivity value as well as half-power beamwidth based on full-wave simulation and *NeF-Net* based approaches. Also, Table 8 summarizes the RMSE error for co- and cross-pol directivity patterns in the two principal planes. In this table, directivity values were normalized between 0 and 1 before calculating RMSE. Notably, both the maximum total directivity and the radiation pattern predicted by the *NeF-Net* are sufficiently close to the full-wave simulations.

Finally, Table 9 presents the prediction time performance and acceleration of our surrogate model compared to the full-wave simulation model. As anticipated, our model demonstrates significantly faster prediction performance (e.g., more than 200 times faster). Although we demonstrate surrogate modeling of a simple microstrip patch antenna that

TABLE 7. Maximum directivity, and half-power beamwidth for the three test designs of Table 5 at resonant frequency.

(a) Maximum directivity (dBi)					
	Full-wave	NeF-Net			
		Prediction (80% data)	Prediction (10% data)	Error (80% data)	Error (10% data)
Antenna 1	7.22	7.0	6.67	3.05%	7.62%
Antenna 2	8.58	8.29	7.44	3.38%	13.29%
Antenna 3	7.46	7.07	6.7	5.23%	10.19%

(b) E-plane half-power beamwidth (degree)					
	Full-wave	NeF-Net			
		Prediction (80% data)	Prediction (10% data)	Error (80% data)	Error (10% data)
Antenna 1	96	97	109	1.04%	13.54%
Antenna 2	66	69	81	4.54%	22.73%
Antenna 3	90	96	104	6.67%	15.55%

(c) H-plane half-power beamwidth (degree)					
	Full-wave	NeF-Net			
		Prediction (80% data)	Prediction (10% data)	Error (80% data)	Error (10% data)
Antenna 1	64	68	73	6.25%	14.06%
Antenna 2	61	63	67	3.28%	9.84%
Antenna 3	63	66	71	4.76%	12.7%

TABLE 8. Far-field prediction RMSE at two principal planes for the three test designs.

		E-plane		H-plane	
		CP	XP	CP	XP
Antenna 1	80% data	0.0478	0.0464	0.1506	0.0338
	10% data	0.0823	0.0278	0.1247	0.0584
Antenna 2	80% data	0.0653	0.0131	0.1732	0.0287
	10% data	0.1158	0.0054	0.2818	0.0989
Antenna 3	80% data	0.1173	0.0038	0.1279	0.04
	10% data	0.1133	0.0046	0.5302	0.0681

TABLE 9. Comparison of time requirements for full-wave simulation and proposed ANN-based estimation in analyzing three test antennas of Table 5 at 51 frequency points.

	Simulation time (in seconds)	ANN-estimation time ^a (in seconds)	Speedup factor
Antenna 1	82	0.36	231
Antenna 2	104	0.37	281
Antenna 3	143	0.36	397

^a Excluding the training time.

can be solved in a reasonable time (e.g., under 2 minutes) with most numerical techniques, we want to highlight the speedup factor achieved with the proposed model. This approach has

great potential to significantly reduce the simulation time for scenarios that require full-wave simulation of large structures, models with lossy materials, or repetitive simulations of similar antenna geometries, such as automated optimization. Also, our results clearly highlight the advantages of saving and reusing datasets for surrogate modeling in such cases.

IV. CONCLUSION

This work investigated the use of convolution neural networks for building a generalized surrogate model that can accurately and very efficiently predict antenna performance parameters. Notably, we focused on applications where a considerable decrease in simulation time is required and prior data is available for deep learning use. Namely, for such applications, we presented *NeF-Net*, a near-field based approach that can accurately predict far-field properties, such as radiation pattern and all its characteristics (e.g., co- and cross-polarization levels, etc.). To demonstrate the efficiency of our method, the widely used rectangular path antenna was utilized. Our *NeF-Net* demonstrated the ability to predict all the necessary antenna parameters, such as, resonant frequency, radiation pattern, directivity, etc., with remarkable accuracy, e.g., 0.0174 average RMSE, and speed, e.g., 200 times faster than conventional full-wave simulations. Future research efforts can focus on further investigating the robustness of this approach, e.g., generalization capabilities across different antenna shapes, frequency bands, etc. Also, the impact of various neural network hyper-parameters, such as the number of training epochs, the number of neurons, etc., on training can be studied for a more efficient implementation of this approach. Nevertheless, the results of this study demonstrate the potential of deep learning techniques in building versatile and ultra-fast electromagnetic estimators that can effectively tackle complex antenna surrogate modeling problems.

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